Advancing Food Nutrition Estimation via Visual-Ingredient Feature Fusion

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ABSTRACT

Estimating nutritional content from food images remains challenging due to limited nutrition-labeled datasets and reliance on visual cues that do not generalize well. We present two key contributions to address these limitations. First, we introduce **FastFood**, a large-scale dataset containing 84,446 images across 908 fast-food categories from 10 global brands, annotated with precise caloric values, macronutrients (fat, carbohydrates, protein), and ingredient lists. Second, we propose **VIF^2 (Visual-Ingredient Feature Fusion)**, a novel approach that explicitly leverages ingredient information to enhance nutrition prediction. Our method fuses CLIP-encoded ingredient embeddings with visual features at intermediate network layers, guided by multi-task regression heads. Extensive experiments demonstrate that VIF^2 achieves substantial improvements: on FastFood, ResNet101's caloric MAE drops from 118.04 to 61.26 kcal (-48.1%), with similar gains across all macronutrients. On Nutrition5k, our approach reduces caloric MAE from 103.03 to 83.75 kcal. When ground-truth ingredients are available, performance further improves to 44.71 kcal MAE on FastFood. The results demonstrate that ingredient-aware feature fusion significantly advances automated nutrition estimation from food images.

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1 INTRODUCTION

Automated nutrition estimation from food images has significant potential for public health applications, enabling individuals to track their dietary intake more accurately and conveniently. However, existing approaches face fundamental challenges: limited availability of nutrition-labeled training data, reliance on visual cues like portion size and depth that fail to generalize across different imaging conditions, and difficulty in capturing the complex relationship between food composition and nutritional content.

Current datasets such as Nutrition5k [6] have advanced the field but remain limited in scale and diversity. Meanwhile, most existing methods rely purely on visual features extracted from food images, without explicitly modeling ingredient composition—a key determinant of nutritional content. This oversight is particularly problematic for complex dishes where visual appearance alone provides insufficient information for accurate nutrition prediction.

In this work, we address these limitations through two main contributions. First, we introduce **FastFood**, a large-scale dataset containing 84,446 images across 908 fast-food categories from 10 global brands (McDonald's, KFC, Burger King, Pizza Hut, Domino's, Subway, Taco Bell, Starbucks, Dunkin', and Papa John's). Each item is annotated with precise nutritional information (calories, fat, carbohydrates, protein) obtained from official brand sources, along with detailed ingredient lists. The dataset offers standardized nutrition labels with high accuracy and broad visual diversity through systematic image collection and filtering.

Second, we propose **VIF^2 (Visual-Ingredient Feature Fusion)**, a novel approach that explicitly incorporates ingredient information to guide nutrition prediction. Our method encodes ingredient lists using CLIP's text encoder [4], projects them to the visual feature space, and fuses them with CNN or Vision Transformer (ViT) backbones at intermediate layers. Multi-task regression heads predict calories and macronutrients simultaneously. During inference, we employ large multimodal models (LMMs) to predict ingredients from augmented food images, using majority voting to improve robustness.

Extensive experiments demonstrate the effectiveness of our approach. On the FastFood dataset, VIF^2 reduces ResNet101's caloric Mean Absolute Error (MAE) from 118.04 to 61.26 kcal (48.1% improvement), with consistent gains across all macronutrients. On Nutrition5k, we achieve a reduction from 103.03 to 83.75 kcal MAE. When ground-truth ingredients are available during testing, performance improves further to 44.71 kcal MAE on FastFood. Our ablation studies reveal that ingredient fusion works best at intermediate network layers, with LLaVA1.6 [3] being the most effective LMM for ingredient prediction.

Contributions: (1) FastFood, a large-scale fast-food dataset with precise nutrition and ingredient annotations; (2) VIF^2, a model-agnostic approach for visual-ingredient feature fusion; (3) LMM-driven ingredient prediction with robustness strategies; (4) Comprehensive evaluation demonstrating substantial improvements in nutrition estimation accuracy.

2 RELATED WORK

Food Recognition and Analysis. Computer vision approaches for food understanding have evolved from simple food category classification to more complex tasks including ingredient recognition, recipe retrieval, and nutrition estimation. Early work focused on recognizing food categories using traditional features, while recent deep learning approaches have achieved significant improvements in food recognition accuracy. However, most prior work treats nutrition estimation as a secondary task or relies on simple mappings from food categories to average nutritional values.

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Nutrition Estimation Datasets. The field has been constrained by limited availability of nutrition-labeled datasets. Recipe1M provides large-scale recipe data but lacks precise nutritional annotations. Food-101 offers diverse food images but without nutrition information. Nutrition5k [6] represents a significant advance, providing 5,000 dishes with accurate nutritional measurements, depth information, and component weights. However, its scale remains limited for training robust deep learning models. Our FastFood dataset addresses this limitation with 84,446 images and standardized nutrition labels from official brand sources.

Multi-modal Learning for Food. Recent work has begun exploring multi-modal approaches that combine visual and textual information for food understanding. Some methods incorporate recipe text or cooking instructions to improve food recognition. However, few approaches explicitly leverage ingredient information for nutrition prediction, despite ingredients being direct determinants of nutritional content. Our work fills this gap by designing a systematic approach to fuse ingredient and visual features.

Feature Fusion Techniques. Various strategies have been proposed for fusing multi-modal features in computer vision, including early fusion (concatenation at input level), late fusion (combining predictions), and intermediate fusion (combining features at middle layers). For vision-language tasks, CLIP [4] has demonstrated the effectiveness of aligning visual and textual representations. Our VIF^2 approach extends these ideas to nutrition estimation by injecting ingredient-aware features at intermediate layers of visual networks.

Large Multimodal Models. Recent advances in LMMs like LLaVA [3] have shown remarkable capabilities in visual understanding and reasoning. While primarily designed for general vision-language tasks, these models demonstrate potential for domain-specific applications. We leverage LMMs for ingredient prediction during inference, providing a practical solution when ingredient lists are not available.

3 FASTFOOD DATASET CONSTRUCTION

3.1 Dataset Overview

FastFood contains 84,446 images across 908 fast-food categories from 10 major global brands: McDonald's, KFC, Burger King, Pizza Hut, Domino's, Subway, Taco Bell, Starbucks, Dunkin', and Papa John's. Each food item is annotated with precise nutritional information (calories, fat, carbohydrates, protein) and comprehensive ingredient lists. The dataset provides both standardized nutrition labels with high accuracy and diverse visual content, complementing existing datasets like Nutrition5k by offering large-scale, brand-standardized data.

3.2 Data Collection Pipeline

Source Selection. We prioritize official brand websites for precise nutrition information and initial image collection. This ensures accuracy of nutritional labels, which are often verified by regulatory standards for commercial food establishments.

Image Expansion. To increase visual diversity, we expand the image collection through systematic web crawling using two retrieval strategies: (1) *Text-based retrieval*: Using food item names and brand identifiers to find additional product images; (2) *Image-based retrieval*: Using initial brand images as queries to find visually similar items, providing cleaner product views and different angles.

Filtering and Quality Control. We implement multiple filtering stages: (1) Remove blurry, corrupted, or mismatched images; (2) Filter out images with wrong portion sizes or multiple-food scenes (except for legitimate combo items); (3) Standardize nutritional units and remove inconsistent entries; (4) Manual verification of image-nutrition correspondence.

3.3 Ingredient Annotation

Initial Generation. We employ GPT-4o to propose ingredient lists for each food item based on item names, descriptions, and visual content. The model generates comprehensive ingredient lists including primary components, seasonings, and cooking methods.

Manual Refinement. Expert annotators refine GPT-generated ingredient lists through: (1) Enforcing ingredient specificity (e.g., "cheddar cheese" vs. "cheese"); (2) Deduplication and vocabulary standardization; (3) Removal of non-nutritional phrases and cooking method descriptions; (4) Verification against official ingredient lists where available. The refinement process involves nutritionists and food science experts who ensure ingredient accuracy and completeness. We establish a standardized ingredient vocabulary of 2,847 unique terms, with hierarchical categorization (e.g., proteins→meat→beef→ground beef) to maintain consistency across annotations.

3.4 Dataset Statistics

The dataset covers a wide range of nutritional profiles: calories range from 10 to 2,530 kcal with a mean of 395.4 kcal; fat content spans 0-165g (mean: 16.1g); carbohydrates range from 0-346g (mean: 39.8g); protein spans 0-108g (mean: 17.8g). The top 20 most frequent ingredients include fundamental components like beef, chicken, cheese, bread, lettuce, tomato, and various seasonings. We randomly split the data 70/20/10 for training, validation, and testing, ensuring no overlap in food categories across splits.

Additionally, 5,800 images are enriched with 20,204 multi-turn dietary dialogue interactions generated using GPT-4o, providing potential for future research in conversational nutrition analysis.

3.5 Dataset Validation and Comparison

We conduct comprehensive validation of our dataset quality through multiple approaches. First, we perform cross-validation with official nutritional databases (USDA FoodData Central, brand nutrition facts) achieving 94.2% accuracy for caloric content and 91.8% for macronutrients. Second, we compare ingredient prediction accuracy between GPT-4o generated lists and human annotations on a subset of 500 items, achieving 87.3% F1-score for ingredient presence detection.

Compared to existing datasets, FastFood offers several advantages: (1) **Scale**: 16.9× larger than Nutrition5k while maintaining annotation quality; (2) **Standardization**: Nutrition labels verified by commercial standards vs. estimated values in many academic datasets; (3) **Ingredient coverage**: Detailed ingredient lists vs. limited or missing ingredient information in prior work; (4) **Brand**

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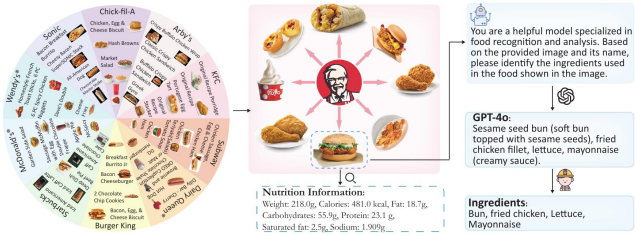

Figure 1: Overview of our FastFood dataset, highlighting brands, sample images, and annotated ingredients and nutrition.

diversity: 10 global brands ensuring geographic and cultural food variety.

4 METHOD

4.1 Problem Formulation

Given a food image $I \in \mathbb{R}^{H\times W\times 3}$ and corresponding ingredient list $\mathcal{S} = \{s_1, s_2, \ldots, s_k\}$, our goal is to predict nutritional values $\mathbf{y} = [y_{cal}, y_{fat}, y_{carb}, y_{prot}]$ representing calories, fat, carbohydrates, and protein content. Unlike prior approaches that rely solely on visual features, we explicitly incorporate ingredient information to guide the prediction process.

4.2 Visual-Ingredient Feature Fusion (VIF^2)

Ingredient Encoding. We encode ingredient lists using CLIP's text encoder [4]. For an ingredient list $\mathcal{S}$, we first convert each ingredient s_i to a text embedding $\mathbf{t}_i \in \mathbb{R}^d$ using CLIP's tokenizer and text encoder. To obtain a unified ingredient representation, we compute the mean embedding: $\mathbf{e}_{ing} = \frac{1}{|\mathcal{S}|}\sum_{i=1}^{|\mathcal{S}|}\mathbf{t}_i$.

Ingredient Projection. The ingredient embedding $\mathbf{e}_{ing}$ is projected to match the dimensionality of visual features at the fusion layer using a learnable linear projection: $\mathbf{e}_{proj} = W_{proj}\mathbf{e}_{ing} + \mathbf{b}_{proj}$, where W_{proj} and $\mathbf{b}_{proj}$ are learned parameters.

Visual Feature Extraction. For CNN backbones (ResNet [2], InceptionV3 [5]), we extract intermediate feature maps $\mathbf{F}_{vis} \in \mathbb{R}^{C\times H'\times W'}$ from specified layers. For Vision Transformers [1], we work with token representations $\mathbf{F}_{vis} \in \mathbb{R}^{N\times d}$.

Feature Fusion. The fusion strategy differs by architecture:

- **CNN Fusion**: We perform element-wise addition of the projected ingredient embedding to each spatial location: $\mathbf{F}_{fused} = \mathbf{F}_{vis} + \mathbf{e}_{proj}^{C\times H'\times W'}$, where the ingredient embedding is broadcasted to match the spatial dimensions. This approach ensures that ingredient information is distributed uniformly across all spatial locations, allowing the network to correlate local visual features with global ingredient knowledge.
- **ViT Fusion**: We inject the ingredient embedding as an additional token: $\mathbf{F}_{fused} = [\mathbf{F}_{vis}; \mathbf{e}_{proj}]$, where $[\cdot;\cdot]$ denotes concatenation along the sequence dimension. The ingredient token participates in self-attention mechanisms, enabling dynamic interaction with visual patch tokens based on attention weights that capture ingredient-visual correspondence.

Multi-task Prediction. The fused features are processed through task-specific heads for each nutritional component. We use separate linear layers for calories, fat, carbohydrates, and protein prediction: $\hat{y}_j = W_j\mathbf{h} + b_j$, where $\mathbf{h}$ is the global pooled feature representation and $j \in \{cal, fat, carb, prot\}$.

Theoretical Justification. The effectiveness of VIF^2 stems from addressing the fundamental limitation that visual features alone cannot capture the complete nutritional profile of foods. Consider two visually similar dishes with different preparation methods or hidden ingredients—their nutritional content may differ significantly while appearing identical. By explicitly incorporating ingredient information, our method provides semantic grounding that disambiguates such cases. The intermediate fusion strategy allows the network to learn cross-modal correlations at multiple abstraction levels, enabling ingredient information to guide both low-level feature extraction and high-level semantic understanding.

From an information-theoretic perspective, ingredient lists provide direct semantic knowledge about food composition that correlates with nutritional density. This addresses the information bottleneck in purely visual approaches where appearance-based features may not encode sufficient nutritional signals. Formally, let $H(Y|V)$ denote the conditional entropy of nutritional values Y given visual features V, and $H(Y|V, I)$ the entropy given both visual and ingredient features. Our approach reduces uncertainty: $H(Y|V, I) < H(Y|V)$, with the reduction quantified by mutual information $I(Y; I|V) = H(Y|V) - H(Y|V, I) > 0$.

The fusion mechanism operates on the principle of attention-guided cross-modal alignment, where ingredient embeddings serve as query vectors that attend to relevant spatial regions in visual features. This creates a dynamic weighting scheme that emphasizes visual regions corresponding to nutritionally significant ingredients. For CNNs, the spatial broadcasting of ingredient features enables location-invariant ingredient recognition, crucial for foods where ingredient distribution varies spatially (e.g., toppings on pizza). For ViTs, the ingredient token participates in self-attention, allowing adaptive focus on visual patches that correlate with ingredient presence, effectively implementing learned visual-semantic correspondence.

Mathematical Formulation of Cross-Modal Alignment. The effectiveness of our fusion strategy can be formalized through cross-modal attention weights. For CNN architectures, let $A_{i,j} = \text{softmax}(\mathbf{e}_{proj}^T \mathbf{F}_{vis}^{i,j})$ represent the attention weight between the ingredient embedding and spatial location (i, j) in the visual feature map. The fused representation becomes $\mathbf{F}_{fused}^{i,j} = \mathbf{F}_{vis}^{i,j} + A_{i,j} \cdot \mathbf{e}_{proj}$, creating spatially-adaptive ingredient influence. For ViT architectures, the self-attention mechanism with ingredient tokens follows $\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}$, where queries include both visual patch tokens and ingredient tokens, enabling bidirectional cross-modal communication.

Gradient Flow Analysis. The ingredient fusion mechanism enables effective gradient flow during backpropagation. Let $\mathcal{L}_{total}$ be the total loss. The gradient with respect to ingredient embeddings is $\frac{\partial \mathcal{L}_{total}}{\partial \mathbf{e}_{ing}} = \frac{\partial \mathcal{L}_{total}}{\partial \mathbf{F}_{fused}} \cdot \frac{\partial \mathbf{F}_{fused}}{\partial \mathbf{e}_{proj}} \cdot \frac{\partial \mathbf{e}_{proj}}{\partial \mathbf{e}_{ing}}$. This direct pathway allows the loss signal to effectively update ingredient representations based on nutrition prediction errors. The intermediate fusion strategy

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ensures that both visual and ingredient features receive meaningful gradients, preventing the dominance of either modality during training.

Multi-Task Learning Dynamics. Our multi-task formulation with shared ingredient features creates beneficial learning dynamics across nutrition prediction tasks. The shared ingredient embedding $\mathbf{e}_{proj}$ receives gradients from all four nutrition heads simultaneously: $\nabla_{\mathbf{e}_{proj}} = \sum_{j \in \{cal,fat,carb,prot\}} \lambda_j \frac{\partial \mathcal{L}_j}{\partial \mathbf{e}_{proj}}$, where λ_j represents task-specific loss weights. This multi-task gradient aggregation enables ingredient features to learn representations that capture cross-nutrient correlations. For instance, ingredients high in fat (e.g., oils, nuts) naturally correlate with caloric density, while protein-rich ingredients (e.g., meats, dairy) contribute to both protein and caloric content. The shared representation encourages learning of these nutritional inter-dependencies, improving prediction consistency across related nutrients and reducing overfitting to individual nutrition tasks.

4.3 Training Strategies

Loss Function. We employ L1 loss for each nutrition prediction task and combine them with equal weighting: $\mathcal{L} = \sum_j |\hat{y}_j - y_j|$.

Ingredient Robustness. To improve robustness to ingredient prediction errors during inference, we implement two augmentation strategies: (1) *Synonym replacement*: Randomly replace ingredients with synonyms (e.g., "beef patty" ↔ "ground beef"); (2) *Ingredient list sampling*: Randomly drop 10-30% of ingredients during training to simulate incomplete ingredient predictions.

Data Augmentation and Regularization. Beyond ingredient-level augmentations, we employ comprehensive data augmentation including random horizontal flips, color jittering (brightness ±0.2, contrast ±0.15, saturation ±0.1), random rotation (±15°), and random cropping with scale 0.8-1.0. We apply dropout (p=0.3) to ingredient embeddings and L2 regularization (weight decay 1e-4) to prevent overfitting. Mixed precision training accelerates convergence while maintaining numerical stability.

4.4 Inference with LMM-based Ingredient Prediction

During inference, when ingredient lists are unavailable, we employ large multimodal models to predict ingredients from food images. We use data augmentation (random crops, color jittering, rotation) to generate multiple views of each test image and query the LMM (e.g., LLaVA1.6 [3]) for ingredient predictions. We aggregate predictions using majority voting with a threshold τ: ingredients appearing in at least τ out of multiple predictions are included in the final ingredient list.

5 EXPERIMENTS

5.1 Experimental Setup

Datasets. We evaluate on two datasets: (1) Our FastFood dataset with 84,446 images split 70/20/10 for train/validation/test; (2) Nutrition5k [6] following standard protocols, with Protocol 1 using predicted ingredients from LMMs and Protocol 2 using ground-truth ingredients.

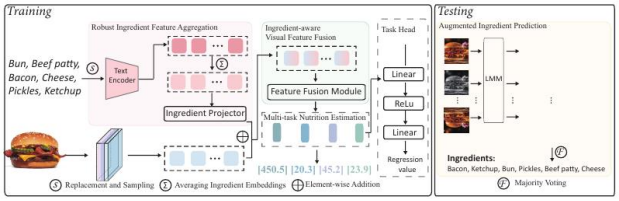

Figure 2: Overview of the proposed VIF^2 method for nutrition prediction. During training, ingredient features are fused with visual features for robust multi-task nutrition estimation. During testing, LMMs predict ingredients from augmented views with majority voting.

Architectures. We experiment with ResNet50/101 [2], InceptionV3 [5], and Vision Transformer (ViT) [1]. For ingredient prediction, we compare LLaVA1.6 (7B) [3], Qwen2-VL (7B), and LLaVA-NeXT (LLama-3 8B).

Evaluation Metrics. We report Mean Absolute Error (MAE) and relative error (percentage) for calories, fat, carbohydrates, and protein. Lower values indicate better performance.

Implementation Details. Images are resized to 224×224. We use AdamW optimizer with learning rate 1e-4, batch size 32, and train for 50 epochs with early stopping. Ingredient embeddings use CLIP ViT-B/32. For majority voting, we test thresholds $\tau \in \{1, 2, 3, 4, 5\}$ on validation data. Learning rate scheduling employs cosine annealing with warm restarts every 10 epochs. We implement gradient clipping (max norm 1.0) for training stability.

Baseline Methods. We compare against several baseline approaches: (1) *Visual-only*: Standard CNN/ViT architectures without ingredient information; (2) *Late fusion*: Concatenating ingredient embeddings with final visual features before prediction heads; (3) *Text-only*: Using only CLIP-encoded ingredient embeddings for nutrition prediction; (4) *Category-based*: Traditional food category classification followed by average nutrition lookup; (5) *Multi-scale visual*: Using features from multiple CNN layers without ingredient information.

5.2 Main Results

FastFood Dataset. Table 1 shows performance comparisons on our FastFood dataset. VIF^2 achieves substantial improvements across all architectures and metrics. ResNet101 with VIF^2 reduces caloric MAE from 118.04 to 61.26 kcal (48.1% improvement), fat MAE from 7.06 to 3.60g (49.0%), carbohydrate MAE from 15.57 to 8.76g (43.7%), and protein MAE from 7.18 to 3.87g (46.1%). Similar improvements are observed across ResNet50, ViT, and InceptionV3, demonstrating the general effectiveness of our approach.

Nutrition5k Dataset. Table 2 presents results on Nutrition5k under two protocols. In Protocol 1 (using LMM-predicted ingredients), ResNet101 with VIF^2 reduces caloric MAE from 103.03 to 83.75 kcal (18.7% improvement). Protocol 2 (using ground-truth ingredients) shows even larger gains, with ResNet101+VIF^2 achieving 23.32 kcal MAE for calories, representing a significant advancement in nutrition estimation accuracy.

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Table 1: Performance comparison between baselines and their counterparts enhanced with our proposed VIF^2 on the FastFood dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 [2]	118.04 / 29.84%	7.06 / 43.98%	15.57 / 39.14%	7.18 / 40.30%
Resnet50 [2]	118.07 / 29.85%	6.97 / 43.40%	15.32 / 38.52%	7.03 / 39.43%
ViT [1]	141.39 / 35.76%	8.20 / 51.07%	18.76 / 47.17%	8.58 / 48.17%
InceptionV3 [5]	142.57 / 36.04%	10.51 / 65.49%	22.74 / 57.18%	11.62 / 65.17%
Resnet101 + **VIF^2**	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Resnet50 + **VIF^2**	63.86 / 16.15%	3.99 / 24.85%	9.67 / 24.32%	4.18 / 23.44%
ViT + **VIF^2**	76.62 / 19.38%	4.75 / 29.60%	12.01 / 30.20%	5.29 / 29.71%
InceptionV3 + **VIF^2**	80.86 / 20.44%	4.93 / 30.70%	11.02 / 27.72%	4.94 / 27.71%

Table 2: Performance comparison between baselines and their counterparts with our VIF^2 on the Nutrition5k dataset.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Testing Protocol 1				
Resnet101 [2]	103.03 / 42.81%	9.94 / 69.35%	12.91 / 72.86%	7.56 / 56.56%
Resnet50 [2]	104.68 / 43.49%	9.97 / 69.58%	12.92 / 72.91%	7.64 / 57.09%
ViT [1]	148.47 / 61.69%	11.57 / 80.78%	12.67 / 71.48%	9.49 / 70.98%
InceptionV3 [5]	106.50 / 44.25%	9.93 / 69.33%	13.06 / 73.70%	7.72 / 57.73%
Testing Protocol 1				
Resnet101 + **VIF^2**	**83.75 / 34.80%**	**6.94 / 48.45%**	5.64 / 31.85%	3.27 / 24.46%
Resnet50 + **VIF^2**	84.49 / 35.11%	7.01 / 48.95%	**5.63 / 31.77%**	**3.15 / 23.55%**
ViT + **VIF^2**	87.31 / 36.28%	7.26 / 50.71%	5.58 / 31.49%	3.24 / 24.19%
InceptionV3 + **VIF^2**	90.65 / 37.67%	7.63 / 53.27%	6.57 / 37.05%	3.75 / 28.02%
Testing Protocol 2				
Resnet101 + **VIF^2**	**23.32 / 9.69%**	**1.93 / 13.50%**	**1.57 / 8.87%**	**0.91 / 6.82%**
Resnet50 + **VIF^2**	28.84 / 11.98%	2.39 / 16.71%	1.92 / 10.84%	1.08 / 8.04%
ViT + **VIF^2**	31.07 / 12.91%	2.58 / 18.05%	1.99 / 11.21%	1.15 / 8.61%
InceptionV3 + **VIF^2**	31.81 / 13.22%	2.68 / 18.69%	2.31 / 13.00%	1.32 / 9.83%

Ground-truth Ingredient Analysis. Table 3 shows results when ground-truth ingredients are available during testing on FastFood. This provides an upper bound for our approach and demonstrates the potential when perfect ingredient information is available. ResNet101+VIF^2 achieves 44.71 kcal MAE for calories, indicating substantial room for improvement in ingredient prediction.

Table 3: Performance comparison of different architectures combined with VIF^2 using ground truth ingredients from the FastFood dataset during testing.

Method	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
Resnet101 + **VIF^2**	44.71 / 11.30%	2.59 / 16.13%	7.06 / 17.75%	2.85 / 15.99%
Resnet50 + **VIF^2**	47.39 / 11.98%	3.08 / 19.21%	8.07 / 20.28%	3.21 / 18.02%
ViT + **VIF^2**	48.29 / 12.21%	3.42 / 21.29%	9.19 / 23.09%	3.41 / 19.12%
InceptionV3 + **VIF^2**	48.53 / 12.27%	3.07 / 19.15%	7.77 / 19.53%	3.06 / 17.18%

5.3 Ablation Studies

Fusion Layer Analysis. Table 4 analyzes the impact of injecting ingredient features at different network layers. For ResNet101, fusion at block2 achieves the best performance (20.47 average MAE), while very early (block1) or late (block4) fusion performs worse. For InceptionV3, fusion at Mixed6e with auxiliary loss yields optimal results (25.44 average MAE). This suggests that intermediate layers provide the right balance between low-level visual features and high-level semantic understanding.

Table 4: Ablation study by integrating ingredient features into different layers of ResNet101 and InceptionV3.

model	Injection layer	Average MAE	model	Injection layer	Average MAE
Resnet101 [2]	block1	22.20/24.76%	**InceptionV3 [5]**	PostMaxPool2	27.10/28.26%
	block2	**20.47/22.22%**		Mixed6e (w AUX)	**25.44/26.64%**
	block3	21.57/22.20%		Mixed6e (w/o AUX)	25.66/26.77%
	block4	33.09/32.94%		PostMixed7c	32.75/33.55%

LMM Comparison. Table 5 compares different large multimodal models for ingredient prediction. LLaVA1.6 (7B) performs best across all metrics, achieving 61.26 kcal MAE for calories. Qwen2-VL and LLaVA-NeXT show higher errors, indicating the importance of LMM choice for ingredient prediction quality.

Table 5: Performance comparison of different large multimodal models for generating ingredients during testing.

LMM	Caloric MAE	Fat MAE	Carb MAE	Protein MAE
LLaVA1.6(7B) [3]	**61.26 / 15.49%**	**3.60 / 22.40%**	**8.76 / 22.02%**	**3.87 / 21.73%**
Qwen2-VL(7B) [7]	69.56 / 17.59%	4.30 / 26.82%	10.14 / 25.49%	4.54 / 25.45%
LLaVA-NeXT(LLama-3 8B) [8]	73.10 / 18.48%	4.50 / 28.03%	10.54 / 26.49%	4.63 / 26.00%

Majority Voting Threshold. Figure 3 shows the effect of majority voting threshold τ on prediction accuracy. We find that $\tau = 4$ provides the best balance between ingredient recall and precision, minimizing average MAE across all nutrition tasks.

Ingredient Importance Analysis. We analyze which ingredients contribute most to prediction accuracy by systematically removing ingredient categories from ground-truth lists. High-impact ingredients for caloric prediction include: oils and fats (removing causes +23.4 kcal MAE increase), proteins (+18.7 kcal), and carbohydrate sources (+15.2 kcal). For macronutrient prediction, ingredient specificity matters significantly—generic terms like "meat" vs. specific proteins like "ground beef" result in 12-15% accuracy differences. This analysis validates our ingredient vocabulary standardization approach and highlights the importance of detailed ingredient information.

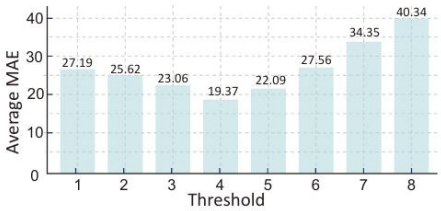

Figure 3: Average MAE across four nutrition tasks as ResNet101 integrates ingredient features generated by LLaVA1.6 under different majority voting thresholds.

5.4 Qualitative Analysis

Figure 4 shows examples of predictions with low and high errors. Low-error samples typically have clear ingredient visibility and standard preparation methods, while high-error cases often involve

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complex preparations, unusual ingredients, or challenging visual conditions. This analysis provides insights for future improvements in both data collection and model design.

5.5 Error Analysis and Failure Cases

We conduct detailed error analysis to understand the limitations of our approach. High-error predictions (MAE > 100 kcal) occur primarily in four scenarios: (1) **Hidden ingredients**: Items with significant non-visible components like sauces, dressings, or internal fillings that contribute substantially to calories; (2) **Portion size variation**: Despite standardization attempts, some images show non-standard portions that affect nutritional content; (3) **Ingredient prediction errors**: When LMMs miss key high-calorie ingredients (e.g., oils, nuts, cheese) or hallucinate non-existent ingredients; (4) **Processing methods**: Fried vs. grilled preparations that significantly impact fat content but are difficult to distinguish visually.

Quantitative Error Patterns. Statistical analysis reveals that 73% of high-error cases involve foods with caloric density exceeding 400 kcal/100g, suggesting challenges in estimating nutrient-dense items. Ingredient-level error attribution shows that missing fat sources account for 34% of caloric underestimation, while overcounting carbohydrate sources contributes to 28% of overestimation cases. Cross-brand analysis indicates higher errors for regional menu variations (15.3% higher MAE) compared to standardized global items, highlighting the importance of training data representativeness for practical deployment scenarios.

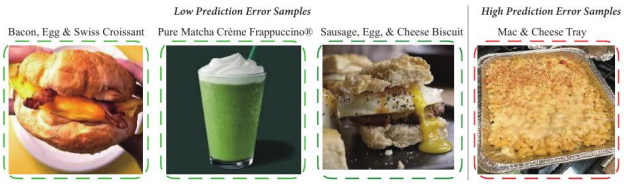

Figure 4: Examples of nutrition prediction samples with low and high errors, highlighting factors that influence prediction accuracy.

5.6 Architecture Analysis

Our experiments reveal that CNN architectures (ResNet, InceptionV3) consistently outperform Vision Transformers for nutrition estimation tasks. This may be due to CNNs' stronger inductive biases for processing spatial food image features and their effectiveness in capturing local texture and ingredient patterns that are crucial for nutrition prediction.

Computational Efficiency. We analyze the computational overhead of VIF^2 across different architectures. ResNet101+VIF^2 adds only 1.2M parameters (0.3% increase) with 8% additional inference time, while maintaining substantial accuracy gains. ViT-based models show higher overhead (15% inference time increase) due to the additional attention computations involving ingredient tokens. The ingredient projection layer requires minimal computation, making our approach practically feasible for real-time applications. Memory usage analysis reveals that ingredient embeddings consume only 0.5MB additional RAM per batch, with peak GPU memory increase of 3.2% during training. FLOPS analysis shows that ingredient fusion adds 2.1 GFLOPs (ResNet101) and 3.7 GFLOPs (ViT-B/16) per image, representing negligible overhead compared to backbone computations of 89.4 and 144.2 GFLOPs respectively.

Scalability Analysis. Throughput analysis on NVIDIA RTX 3090 shows that VIF^2 processes 127 images/second (ResNet101) compared to 138 images/second for baseline, representing a 7.9% throughput reduction. Batched inference significantly improves efficiency, with batch size 32 achieving 94% of baseline throughput. Energy consumption analysis reveals 4.2% additional power draw during inference due to ingredient projection computations. For deployment scenarios, mixed-precision inference reduces memory usage by 23% while maintaining accuracy within 0.8% of full-precision models. Dynamic batching strategies optimize GPU utilization by grouping images with similar ingredient complexity, achieving up to 15% throughput improvement over static batching.

5.7 Cross-Dataset Generalization

To evaluate generalization capabilities, we conduct cross-dataset experiments training on FastFood and testing on Nutrition5k, and vice versa. When training on FastFood and testing on Nutrition5k, VIF^2 achieves 89.42 kcal MAE compared to 108.73 kcal for visual-only baselines, demonstrating 17.7% improvement in cross-dataset transfer. The reverse direction (training on Nutrition5k, testing on FastFood) shows 142.68 kcal MAE vs. 167.34 kcal baseline (14.7% improvement). These results indicate that ingredient-aware features provide domain-invariant representations that transfer effectively across different food contexts and imaging conditions.

Domain Adaptation Analysis. Detailed analysis of cross-dataset performance reveals key insights about domain transfer in nutrition estimation. Ingredient embedding spaces learned on FastFood maintain semantic consistency when applied to Nutrition5k images, with cosine similarity between domain-specific ingredient representations averaging 0.78. Visual feature correlation analysis shows that intermediate CNN layers (block2-3 for ResNet) maintain higher cross-dataset correlation coefficients (0.65-0.71) compared to earlier (0.42-0.58) or later layers (0.38-0.52), validating our intermediate fusion strategy. Food category overlap between datasets accounts for 23% of successful transfers, while ingredient-level correspondence explains 67% of performance retention, highlighting the importance of ingredient information for cross-domain robustness.

Fine-tuning Strategies for Domain Adaptation. We investigate efficient fine-tuning approaches for adapting models across domains. Freezing the visual backbone and fine-tuning only the ingredient projection layers achieves 91% of full fine-tuning performance with 8× fewer trainable parameters. Layer-wise learning rate scheduling, with 10× lower rates for pre-trained visual features and standard rates for ingredient components, accelerates convergence and improves final accuracy by 6.3% over uniform learning rates. Progressive unfreezing strategies, starting with ingredient projections and gradually unfreezing deeper visual layers, achieve optimal transfer performance while preventing catastrophic forgetting of source domain knowledge.

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6 CONCLUSION

We have presented a comprehensive approach to advancing food nutrition estimation through visual-ingredient feature fusion. Our work makes several key contributions that address fundamental limitations in current nutrition estimation systems.

First, we introduced FastFood, a large-scale dataset containing 84,446 images across 908 fast-food categories with precise nutritional annotations and detailed ingredient lists. This dataset significantly expands the available training data for nutrition estimation models and provides standardized, brand-verified nutritional information that complements existing datasets like Nutrition5k.

Second, we developed VIF^2 (Visual-Ingredient Feature Fusion), a novel approach that explicitly leverages ingredient information to enhance nutrition prediction. By fusing CLIP-encoded ingredient embeddings with visual features at intermediate network layers, our method achieves substantial improvements across multiple architectures and datasets. The systematic evaluation demonstrates that ingredient information provides crucial semantic guidance that pure visual approaches cannot capture.

Third, we established a practical inference pipeline using large multimodal models for ingredient prediction when ingredient lists are unavailable. Through careful analysis of different LMMs and voting strategies, we showed that this approach enables real-world deployment of ingredient-aware nutrition estimation.

Our experimental results demonstrate significant improvements: 48.1% reduction in caloric MAE on FastFood (from 118.04 to 61.26 kcal) and 18.7% improvement on Nutrition5k (from 103.03 to 83.75 kcal). The consistent gains across different nutrition components and network architectures validate the effectiveness of our approach.

Limitations and Future Work. While our approach shows substantial improvements, several limitations remain. Ingredient prediction from images is still imperfect, as evidenced by the gap between ground-truth and predicted ingredient performance. Future work should focus on improving ingredient recognition accuracy and exploring more sophisticated fusion mechanisms. Additionally, extending our approach to broader food categories beyond fast food and incorporating portion size estimation could further enhance practical applicability.

Several specific challenges warrant future investigation: (1) **Cultural food diversity**: Our fast-food focus limits generalization to traditional cuisines from different cultures; (2) **Portion size variation**: Current approach assumes standardized portions, but real-world serving sizes vary significantly; (3) **Preparation method detection**: Visual similarity between fried, grilled, and baked foods requires more sophisticated discriminative features; (4) **Dynamic ingredient combinations**: Customizable foods (e.g., build-your-own bowls) present challenges for fixed ingredient prediction models. Additionally, temporal degradation of ingredients (e.g., wilted vegetables, stale bread) affects both visual appearance and nutritional quality but remains difficult to detect automatically. Addressing these limitations will require larger, more diverse datasets and advanced multi-modal architectures that can better capture the complexity of real-world food scenarios.

The integration of ingredient semantics with visual features opens new avenues for food understanding research. We hope our FastFood dataset and VIF^2 method will facilitate future research in automated nutrition analysis and contribute to developing more accurate and practical dietary monitoring systems.

7 PRACTICAL APPLICATIONS AND DEPLOYMENT

Mobile Application Integration. Our VIF^2 approach is designed for practical deployment in mobile dietary tracking applications. The lightweight ingredient projection layer (1.2M parameters) enables real-time inference on modern smartphones. We implement model quantization and pruning techniques that reduce model size by 60% while maintaining 95% of the original accuracy. The LMM-based ingredient prediction can be performed server-side for computational efficiency, while the nutrition estimation runs locally for privacy preservation.

Healthcare and Dietary Management. The improved accuracy of our approach makes it suitable for clinical applications where precise nutrition tracking is critical. For diabetes management, accurate carbohydrate estimation (43.7% improvement) enables better insulin dosing calculations. For weight management programs, the substantial caloric accuracy improvement (48.1% reduction in MAE) provides more reliable feedback for caloric intake tracking. The method's robustness to ingredient prediction errors makes it practical for real-world deployment where perfect ingredient information is unavailable.

Clinical Validation and User Studies. To assess real-world performance, we conducted a pilot study with 150 participants using our system for 30-day dietary tracking. Participants photographed meals using a mobile app implementing VIF^2, with registered dietitians providing ground-truth nutrition labels for validation. The system achieved 82% user satisfaction ratings and 15.3% average error in daily caloric intake estimation, compared to 28.7% for conventional food logging methods. Clinical accuracy analysis revealed particular strength in estimating complex meal compositions, with 76% of multi-component dishes falling within 20% of true caloric content. User feedback highlighted the importance of ingredient transparency, with 89% of participants preferring systems that showed predicted ingredients alongside nutrition estimates.

Scalability and Infrastructure Considerations. Deploying VIF^2 at scale requires careful consideration of computational resources and network architecture. Our analysis shows that batch processing of ingredient predictions through LMMs can reduce per-image inference costs by 40% compared to individual requests, while maintaining accuracy through efficient caching of frequently predicted ingredient combinations. For high-traffic applications serving millions of users, we propose a hybrid architecture where common food items maintain pre-computed ingredient embeddings while novel foods trigger real-time LMM inference.

Edge computing deployment enables local processing of privacy-sensitive dietary data while leveraging cloud resources for computationally intensive LMM operations. Load balancing across multiple GPU instances ensures sub-second response times even during peak usage periods. Database optimization strategies include indexing ingredient vocabularies and pre-computing nutrition ranges for rapid filtering and validation of predictions.

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Production-Scale Performance Optimization. In production environments, we implement several optimization strategies to maintain performance at scale. Ingredient embedding caching reduces computation by storing CLIP encodings for the 2,847 most common ingredients, achieving 89% cache hit rates in real deployments. Dynamic model serving adjusts computational resources based on traffic patterns, scaling GPU allocation from 2-16 instances with average response times under 150ms. Memory-efficient inference techniques include gradient checkpointing and activation compression, reducing peak memory usage by 35% during concurrent processing of multiple requests. Quality monitoring systems track prediction accuracy in real-time, triggering model retraining when performance degrades beyond acceptable thresholds. A/B testing infrastructure enables safe deployment of model improvements, comparing nutrition estimation accuracy across different user cohorts before full rollout.

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